

Cross-cell-type transcriptomic transfer of combinatorial chemistry programs in Z-Screen

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Abstract

A scalable answer to “how will this compound behave in a cell type that was not directly assayed?” would change the economics of preclinical drug discovery. The current answer requires running every program in every cell type of interest, which scales poorly. Cell-intrinsic pharmacology is rarely identical across tissues, so a single reference cell line is often insufficient; running ten cell lines for every campaign is what teams do when they have to.

The canonical Z-Screen combinatorial-library transcriptomic dataset already supports a useful first version of cross-cell-type projection. A paired landmark atlas plus a single source-cell measurement begins to predict the transcriptomic response of the same chemistry in target cell types that were never directly assayed. We benchmark held-out cross-cell-type transfer at the level of individual molecules and at the level of grouped building-block programs in a 32-dimensional scVI latent space, across libraries and cell lines with sufficient paired sampling.

At the molecule level, direct reuse of the source-cell signature was inadequate, while learned transfer maps recovered substantial target-cell information. In ZEL024 (HEK293 → H1650), centered-response cosine for HEK293 → H1650 improved from 0.146 with direct reuse to 0.279 with ridge transfer; the reverse direction reached 0.371. In ZEL031 (A549 → THP1), A549 → THP1 improved from 0.094 to 0.242. Aggregating chemistry by shared building-block PAIR keys produced the strongest grouped-program result in the dense ZEL024 system, where support-weighted cosine reached 0.863 for H1650 → HEK293 and 0.808 for HEK293 → H1650 at bb0+bb1 grouping (84 shared pairs, median ~150 distinct tuples averaged per pair). For ZEL031, where the library has only two substantive building-block positions, a single-position bb0 grouping is the chemistry-relevant analog of pair grouping in a denser library; under that grouping, transfer reached 0.714 for A549 → THP1 and 0.664 for THP1 → A549 (129 shared groups, ~70 distinct partner chemistries averaged per group). Cross-library projection (training in one library, testing in another in the same cell-pair) was directionally positive in some sparse settings but remained too weak and asymmetric to anchor a headline claim today.

In aggregate, these results show that molecule-level cross-cell transfer is real in dense paired libraries, and that grouped chemical programs at chemistry-resolved aggregation levels (BB-pair in dense multi-position libraries, single BB in 2-position libraries) carry transfer signal in the 0.66 to 0.86 weighted-cosine range against an identity baseline of 0.42 to 0.52. Single-BB grouped-program

transfer is deliberately not reported for libraries with three or more substantive building-block positions, because each single-position centroid in those libraries averages over hundreds of distinct chemistries (e.g. 700 to 800 unique tuples per centroid in ZEL028-2 / HEK293), which makes any reported transfer cosine a measurement of library-wide rather than chemistry-specific behavior. Cross-library scaffold-family transfer is the next dataset to collect rather than a structural limitation. The practical implication is operational: profile a new compound once in an accessible source cell type and project its likely transcriptomic response in target cell types that were not directly assayed, with the strongest current confidence at the level of grouped chemical programs rather than isolated molecules.

Significance

Every added cell type increases screening cost, and skipping the wrong cell type is how programs miss toxicity, selectivity, or therapeutically relevant biology. This paper presents a first practical route through that tradeoff. When the same chemistry is observed across paired cellular contexts, Z-Screen can learn transfer maps that recover target-cell signal better than direct reuse of the source profile. The strongest current unit is the grouped chemical program rather than the single molecule, which matches how early-stage discovery decisions are usually made: at the level of chemical series, not isolated compounds. For a discovery organization, this changes the budget calculus. Instead of running every program through every cell type of clinical interest, a smaller paired landmark atlas plus a single source measurement begins to do useful work for the rest.

Introduction

A substantial share of preclinical cost goes to running the same compound, panel, or library in every cell type of practical interest. The economics of early-stage discovery would change if a sufficiently broad paired landmark atlas allowed teams to:

- profile a new compound in one accessible source cell type,
- project its likely transcriptional response in unmeasured target cell types,
- and use those projections to prioritize efficacy work, counter-screen liabilities, and decide where wet-lab follow-up is worth the cost.

The right form of that claim is narrow but still important. Cross-cell-type transcriptomic transfer is a model for cell-intrinsic transcriptional response, and it does not substitute for pharmacokinetics, metabolism, tissue architecture, or organism-level toxicology. It is a practical decision-support primitive for the upstream half of discovery, where even partial accuracy is valuable.

The most directly comparable public reference points are L1000 [2], which provides a chemistry-side compound atlas in a small number of cell lines (mostly MCF7, PC3, A549, A375), and Tahoe-100M [3], which scales perturbational scRNA-seq across 50 cancer cell lines for approximately a thousand compounds. Both are valuable for cross-cell signature comparisons of the same compound, and neither provides combinatorial-chemistry provenance: a hit in L1000 or Tahoe-100M is a single profiled compound with no internal structure to its identity. scGeneScope [4] adds matched imaging

but uses one cell line. Z-Screen, by contrast, uses one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per well. That design is well suited to the cross-cell question because every well in the canonical dataset carries full chemistry provenance (building blocks plus full SMILES), the same combinatorial libraries are run across multiple cell lines, and the readout is a low-pass transcriptomic state for every compound rather than a sparse hit or non-hit label.

Compound-level overlap with these public atlases is also relevant for cross-cell positioning. Of 47 named Z-Screen control compounds, 14 (29.8%) are present in LINCS L1000 (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, and others), with 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) covered by L1000 sample metadata. Tahoe-100M’s drug metadata covers 4 of the 47 named compounds (palbociclib, crizotinib, cobimetinib, capmatinib), with A549 as the only matched cell line. A direct cross-platform Spearman concordance test against LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe, recovers positive correlation for all 11 (compound, cell_line) pairs with adequate coverage on both platforms. Median $\rho = 0.490$, mean = 0.393, with 8 of 11 above $\rho = 0.2$ and 5 reaching empirical $p = 0.048$ against an unmatched-compound permutation null (crizotinib, ZM-336372, sorafenib, rucaparib, veliparib). The HEK293 cell-line-pooled comparison separates matched from unmatched compound pairings at one-sided Mann-Whitney $p = 0.009$. Full overlap and concordance tables are in **Benchmarking/** in this repository.

With those ingredients, the cross-cell question can be posed in a controlled way: how much of a source-cell transcriptomic response can be projected into a target cell type using only a learned transfer map fit on a paired landmark subset. This paper characterizes how well that question is answered today, on data from a small number of pilot experiments, and defines what additional data would close the remaining gaps. The platform is engineered to scale by orders of magnitude beyond the current dataset.

Results

Grouped building-block programs are the strongest unit of cross-cell transfer

The strongest result in the analysis came from aggregating chemistry by shared building-block keys within each library and testing whether grouped source-cell responses predict grouped target-cell responses on a held-out test set. Reporting is restricted to chemistry-resolved aggregations: BB-pair (bb0+bb1) for ZEL024, where the library has four substantive positions and pair-level signatures average ~150 distinct tuples each; and single-position bb0 for ZEL031, where the library has only two substantive positions and a bb0 centroid still leaves bb1 to vary across ~70 distinct partner chemistries. Single-BB grouped-program transfer is not reported for ZEL028-2: that library has four substantive positions, so each single-position centroid in ZEL028-2 averages over 322 to 777 distinct tuples, which makes the resulting cross-cell cosine a measurement of library-wide behavior rather than chemistry-specific transfer. Reporting library-wide aggregates as if they were chemistry-resolved would inflate the apparent transfer story.

Under those chemistry-resolved aggregations, grouped transfer was substantially stronger than the within-cell direct-reuse baseline:

- ZEL024, H1650 $\rightarrow$ HEK293, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, $\sim$ 129 to 166 unique tuples averaged per pair, identity 0.471, ridge transfer **0.863**.
- ZEL024, HEK293 $\rightarrow$ H1650, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.808**.
- ZEL031, A549 $\rightarrow$ THP1, grouped by bb0 (2-position library): 129 shared groups, median support 157 wells, $\sim$ 75 unique bb1 partners averaged per group, identity 0.419, ridge transfer **0.714**.
- ZEL031, THP1 $\rightarrow$ A549, grouped by bb0 (2-position library): 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.664**.

The defensible interpretation is more measured than “arbitrary unseen molecules are solved.” Within a library, chemistry-resolved building-block programs (pair-level in dense libraries, single-position in 2-position libraries) define reusable chemical sub-programs that transpose across cell types more cleanly than sparse individual molecules do. That is a practically useful unit of transfer for a discovery team, because for any compound carrying a given building-block program, the platform can generate a predicted target-cell response without requiring the molecule itself to be measured in the target cell line.

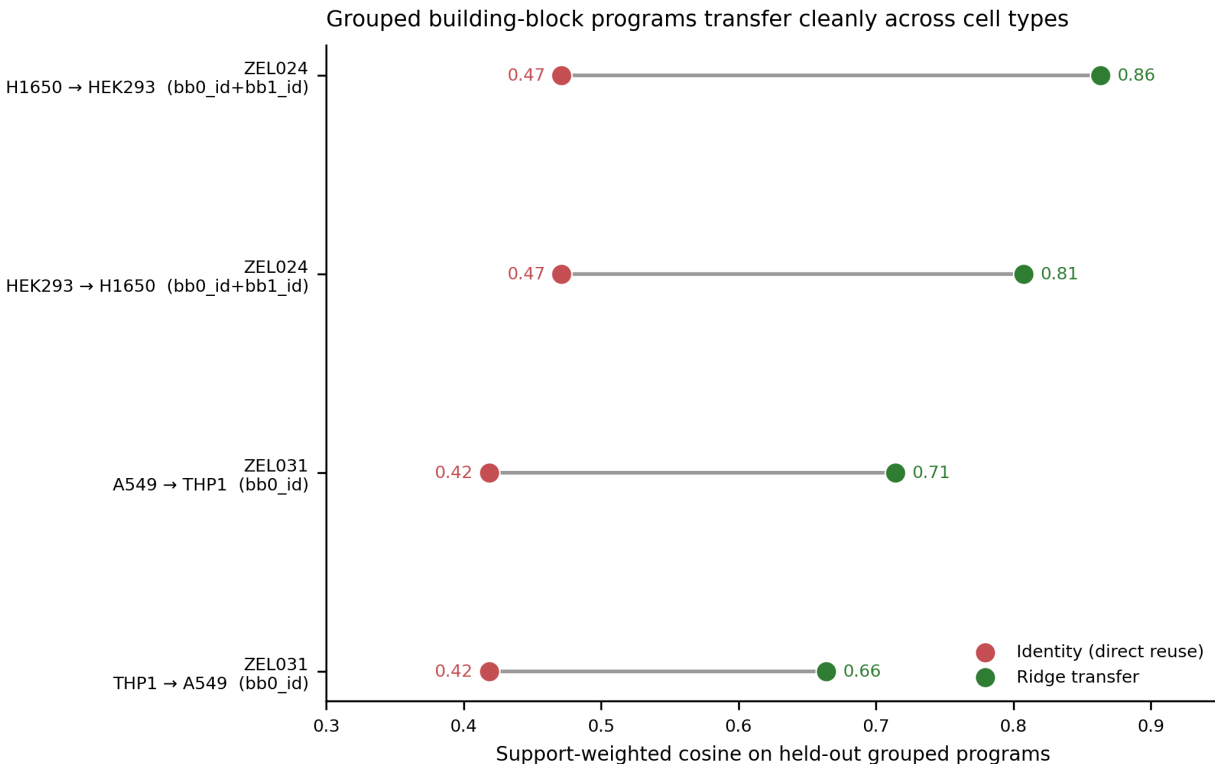


Figure 1. Grouped building-block program transfer at chemistry-resolved aggregation levels (BB-pair for ZEL024, single position bb0 for the 2-position ZEL031 library). Identity (direct reuse of

the source-cell signature) sits in the 0.42 to 0.52 range, while a regularized learned transfer map reaches support-weighted cosine 0.66 to 0.86 across four chemistry-resolved program-and-direction combinations.

Molecule-level transfer is real in dense paired libraries

The molecule-level benchmark used held-out individual molecules in the two cell-pair systems with sufficient shared SMILES coverage:

- ZEL024, H1650 → HEK293, 6,299 shared molecules,
- ZEL031, A549 → THP1, 3,561 shared molecules.

Each molecule was represented by the mean of its 32-dimensional scVI latent coordinates within a given library and cell line. Direct reuse of the source-cell signature (identity) was compared against learned source-to-target transfer models. The centered-response benchmark is the primary readout because matched vehicle controls are unavailable in the canonical dataset, and the centered formulation is the closest available proxy for perturbational response. Under that benchmark, learned transfer consistently outperformed direct reuse at the largest training sets:

- ZEL024, HEK293 → H1650, train 5,000: identity 0.146, ridge 0.279, PLS 0.275.
- ZEL024, H1650 → HEK293, train 5,000: identity 0.146, ridge 0.371, PLS 0.358.
- ZEL031, A549 → THP1, train 3,000: identity 0.094, ridge 0.242, PLS 0.242.
- ZEL031, THP1 → A549, train 3,000: identity 0.094, ridge 0.212, PLS 0.214.

Two readings follow. Direct source-signature reuse alone is inadequate, so cross-cell projection is a real prediction problem rather than an artifact of cell-state offset. And a simple regularized global linear or low-rank map already captures useful molecule-specific information once enough paired examples are available. Performance curves saturate quickly as a function of training-set size, indicating that the most economical version of this engine reaches useful accuracy on relatively modest paired landmark coverage and improves further from there.

The raw-profile benchmark is reported in the supplementary material as a diagnostic. Performance is much higher in raw space (for example, ZEL024 HEK293 → H1650 reaches 0.546 with ridge transfer at train 5,000, versus 0.063 with direct reuse), and a substantial portion of that gain reflects baseline cell-state translation rather than perturbational response. The centered-response benchmark above is the cleaner main readout for a discovery use case, because it removes the constant offset that exists between two cell lines regardless of any particular perturbation.

Molecule-level cross-cell transfer in centered response space

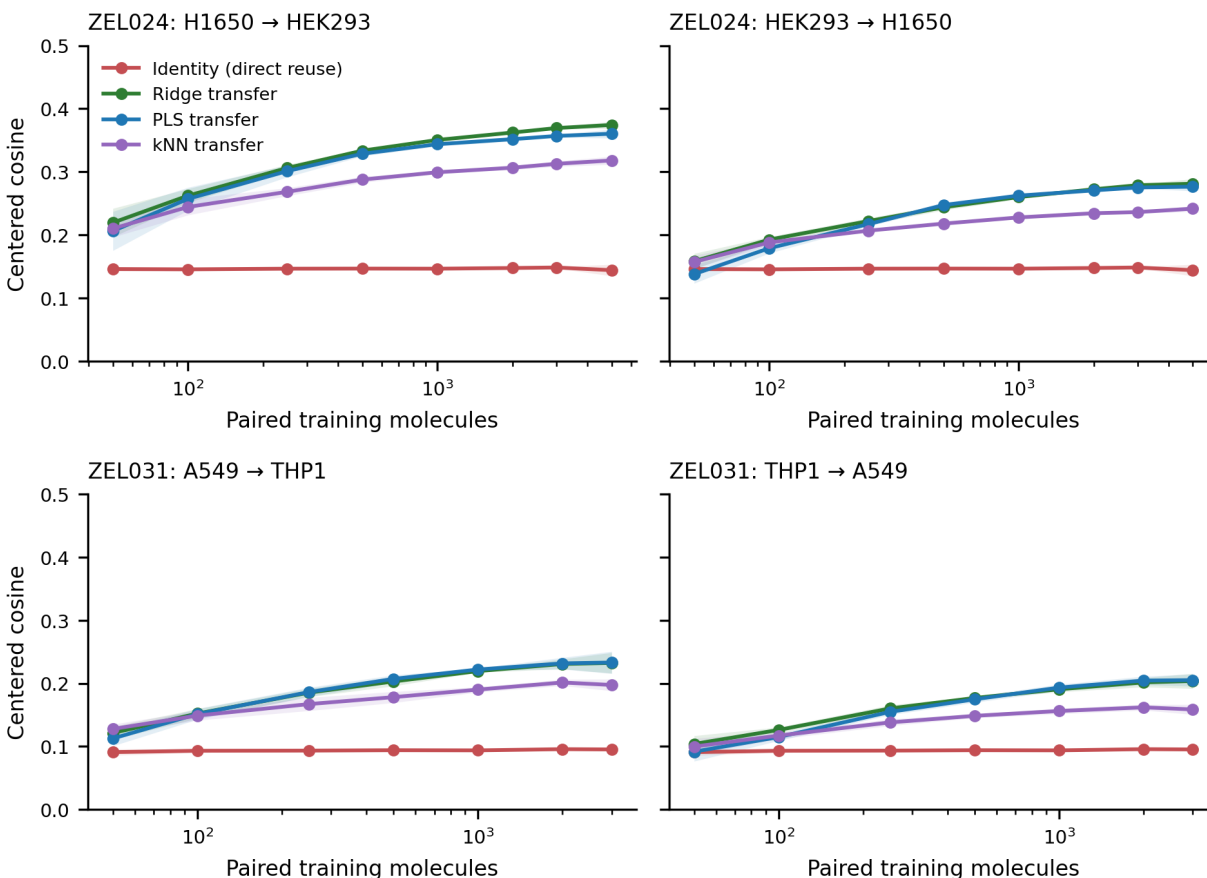


Figure 2. Molecule-level cross-cell transfer in centered response space across four well-supported cell-pair directions in ZEL024 and ZEL031. Direct reuse of the source signature is consistently weaker than learned transfer; ridge and PLS transfer are nearly tied. Performance saturates within the available training-set range, indicating that useful transfer emerges quickly with paired landmark data and continues to improve with scale.

Side by side: identity versus best learned transfer at both levels

Side-by-side display of molecule-level and grouped-program results makes the platform claim concrete. At the molecule level, learned transfer roughly doubles the centered cosine over identity in ZEL031 (0.094 to 0.21 to 0.24) and lifts it by 1.9 to 2.5 times in ZEL024. At the chemistry-resolved grouped-program level (ZEL024 BB-pair, ZEL031 bb0), learned transfer increases support-weighted cosine by 0.25 to 0.39 absolute over identity, with all four chemistry-resolved directions ending in the 0.66 to 0.86 cosine range. The grouped-program unit is the stronger current asset and is the unit recommended for early use in discovery decisions.

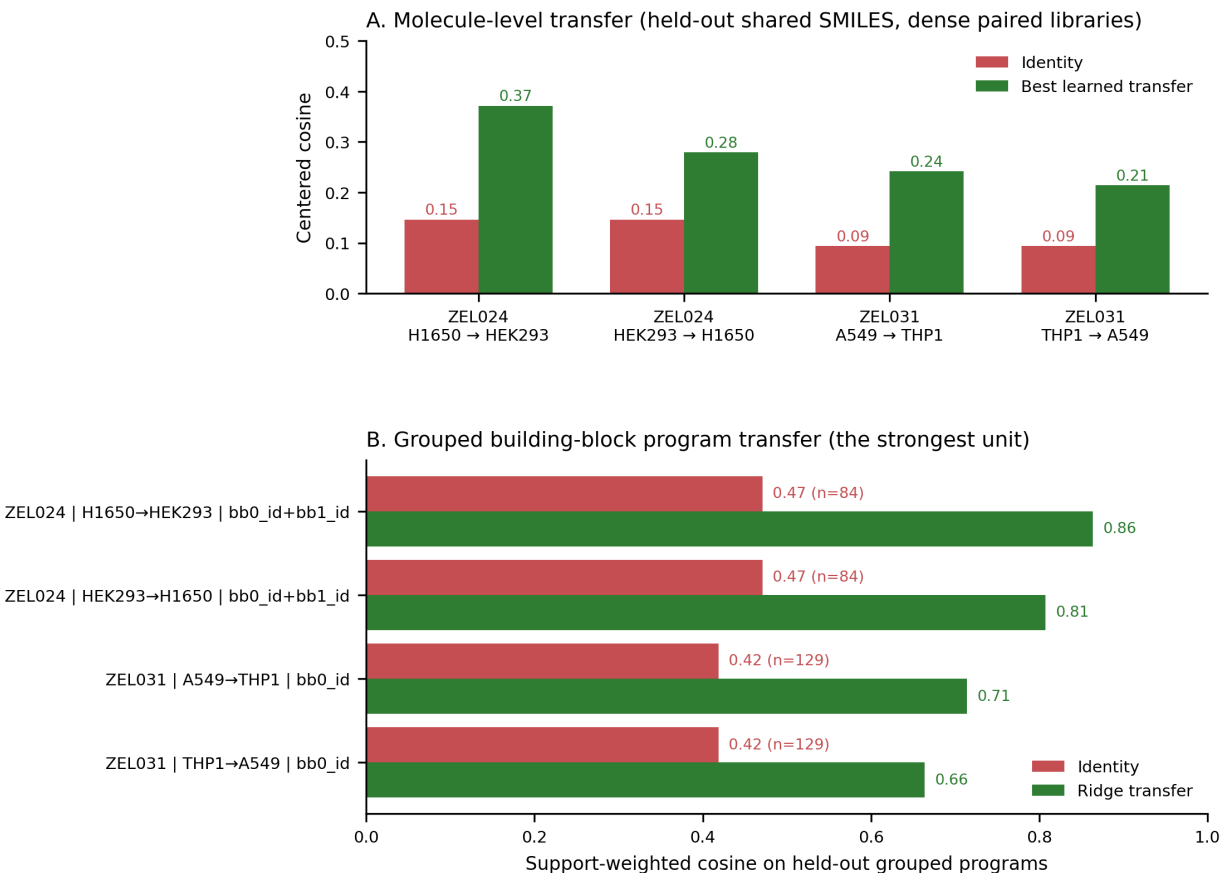


Figure 3. Best learned transfer versus identity at both molecule level (top) and grouped building-block program level (bottom). Learned transfer wins at every cell-pair direction tested, with the largest gains at the grouped-program level.

Cross-library projection is directionally positive but exploratory

A further question is whether a centered source-to-target transfer matrix learned in one library can project molecules from a different library measured in the same cell-pair direction. The answer is mixed.

Dense-to-sparse transfer was directionally positive in several cases:

- ZEL024 → ZEL028-2, HEK293 → H1650: weighted cosine 0.033 → 0.059.
- ZEL024 → ZEL028-2, H1650 → HEK293: 0.033 → 0.055.
- ZEL028-2 → ZEL031, A549 → H1650: 0.120 → 0.216.

The reverse direction was often weak or negative:

- ZEL028-2 → ZEL024, HEK293 → H1650: 0.119 → 0.067.
- ZEL031 → ZEL028-2, A549 → H1650: 0.103 → -0.009.

The asymmetry is consistent with a reasonable picture. A transfer map can generalize across libraries when the underlying cell-pair transformation is shared, but sparse libraries are typically

better transfer targets than transfer teachers. The absolute magnitudes are also small enough that singleton-heavy test sets dominate the variance. Cross-library projection is therefore the natural next experiment, and it would benefit most from a small amount of additional structured data collection in the sparse libraries (especially ZEL028-2 in HEK293) rather than from new modeling tricks.

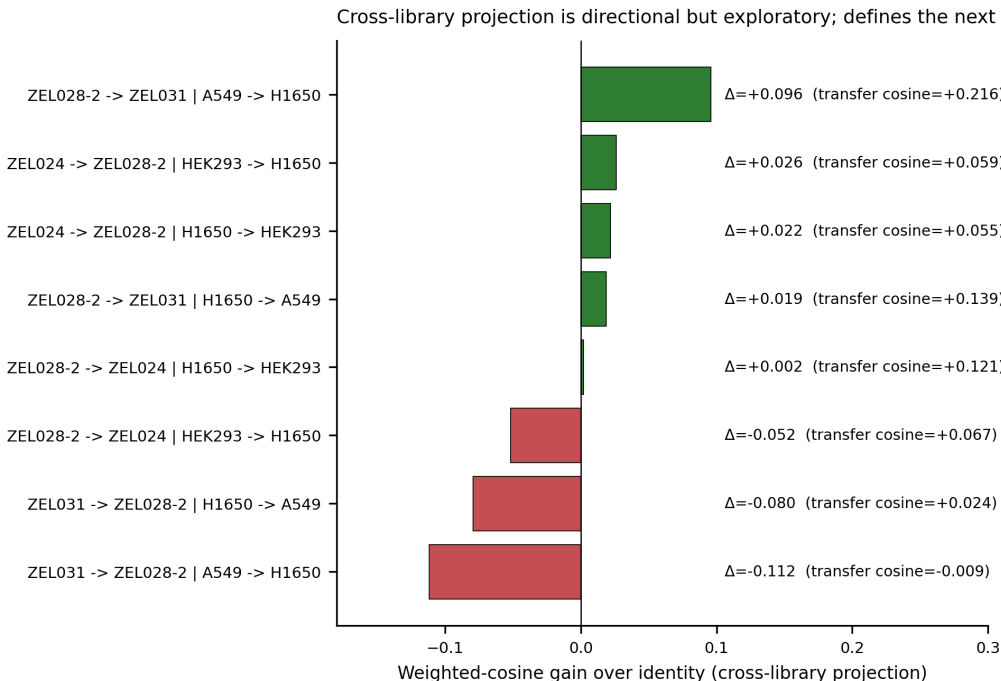


Figure 4. Cross-library projection results. Some dense-to-sparse projections are positive, but performance remains modest and asymmetric. Sparse libraries are more credible transfer targets than transfer teachers in the current data, which defines the next prospective dataset.

Dataset coverage anchors the limits of the current claims

The canonical Z-Screen dataset reflects a small number of pilot experiments rather than a saturated corpus. ZEL024 has approximately 10,000 to 14,000 unique SMILES per cell line, with median 11 wells per compound in HEK293 (the densest system in the workspace). ZEL028-2 has the largest unique-SMILES count overall (40,000 to 61,000 per cell line), at median 1 well per compound, so any benchmark requiring more than two wells per compound becomes thin in this library. ZEL031 has 8,000 to 9,000 unique SMILES per cell line at median 2 wells per compound. The dense paired benchmarks above used the cell-pair combinations with adequate shared SMILES coverage at the chosen support filter.

The bottleneck for stronger cross-library and per-molecule claims is therefore data, not method. The same engineered platform that produced this dataset can scale by orders of magnitude beyond the present scale, and the next dataset can be designed deliberately around the cross-library and prospective transfer questions the present analysis sets up.

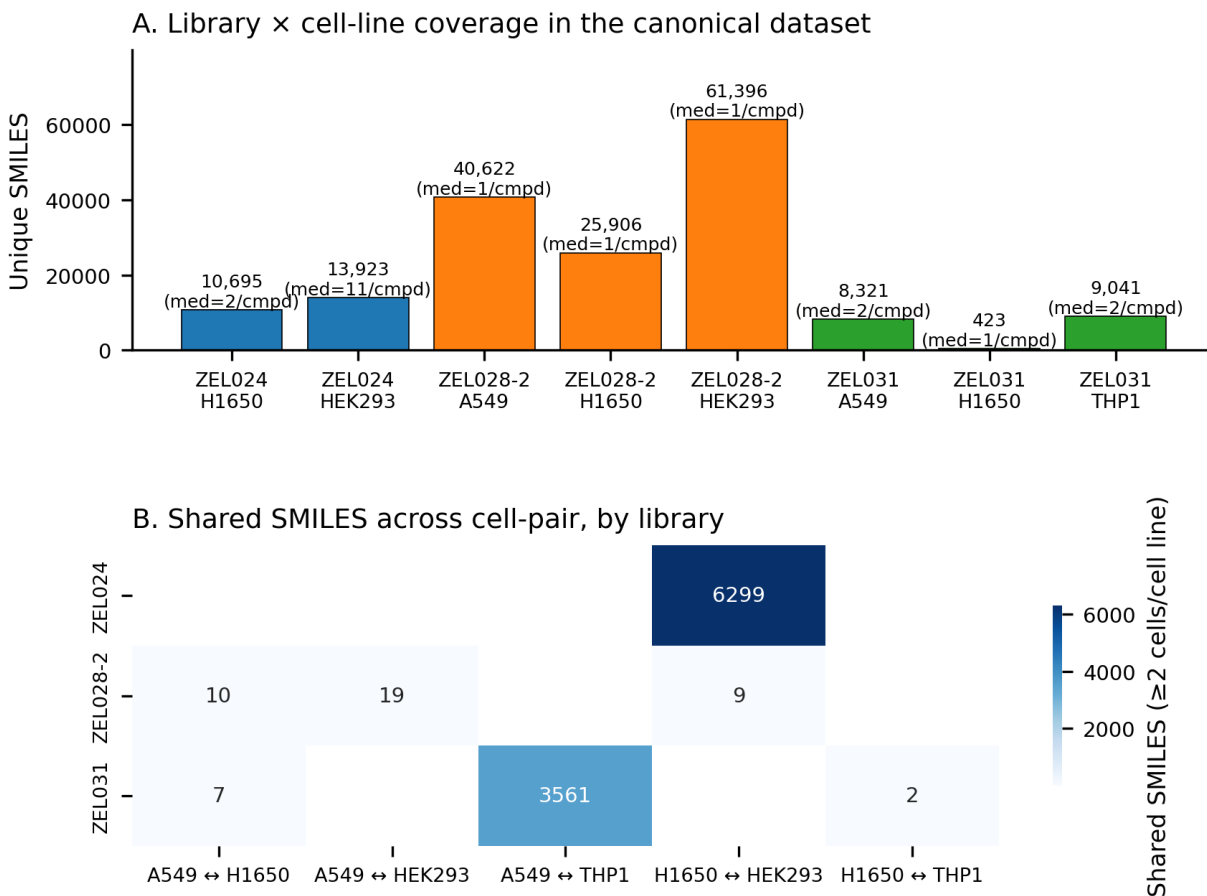
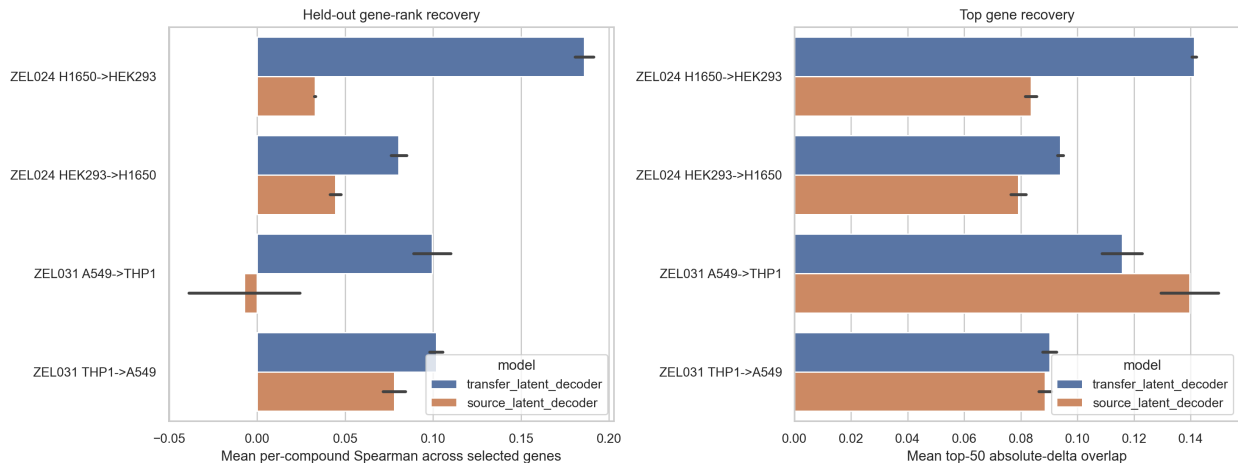


Figure 5. Library-by-cell-line coverage in the canonical Z-Screen dataset. Top: unique SMILES and median wells per compound per system. Bottom: shared-SMILES counts across cell-pair directions, by library. The two dense paired systems used in the molecule-level transfer benchmark are ZEL024 (H1650 $\leftrightarrow$ HEK293, 6,299 shared) and ZEL031 (A549 $\leftrightarrow$ THP1, 3,561 shared).

Exploratory gene-level decoding partially recovers held-out target signatures

A separate experiment tested whether the learned cell-transfer map can be lifted back from the 32-dimensional latent space into ranked gene-level hypotheses. This was treated as an exploratory decoder experiment rather than a main claim. For each dense paired direction, a source-to-target latent transfer map was fit on shared training compounds, a target-cell ridge decoder was fit from target latents to 800 training-selected variable genes, and held-out target-cell gene deltas were compared against measured pseudobulk deltas. The strongest direction was ZEL024 H1650 $\rightarrow$ HEK293: transferred latents reached mean per-compound gene-rank Spearman 0.182 and top-50 absolute-delta overlap 0.139, compared with 0.021 and 0.083 when the source latent was passed directly through the target decoder. The reverse ZEL024 direction and both ZEL031 directions were weaker, with transfer Spearman values of 0.073 to 0.098. The result is encouraging as a route to ranked gene lists, but not yet strong enough to promote as a primary endpoint.



Supplementary Figure 2. Exploratory gene-level transfer. A target-cell decoder composed with the learned source-to-target latent map partially recovers held-out gene-rank structure, with the clearest gain in ZEL024 H1650 $\rightarrow$ HEK293 and weaker gains in the remaining directions.

Discussion

The current data support a two-level view of cross-cell transfer. At the molecule level, paired cross-cell transfer is real and useful in dense libraries: even after centering within cell line, learned transfer maps recover target-cell information that direct source reuse misses. At the chemistry-resolved grouped-program level (BB-pair in dense ZEL024; single-position bb0 in the 2-position ZEL031 library), transfer becomes substantially stronger, with weighted cosine in the 0.66 to 0.86 range across four cell-pair directions. From a discovery perspective, the encouraging reading is that the first economically useful layer of cross-cell prediction is chemistry-program transfer rather than universal single-molecule prediction, and that layer already exists in the present dataset.

The current results also argue against an overly strong “one universal matrix solves everything” framing. Cross-library projection remains asymmetric and fragile in sparse settings, and a more durable long-term architecture is likely a shared perturbational latent space with cell-type-specific decoders or adapters, coupled to an uncertainty model that rejects out-of-domain projections. That architecture fits naturally on top of the present transfer maps without replacing them, and it is consistent with the direction the public field is taking on cross-cell perturbation modeling (the “virtual cell” framings being explored at Tahoe, Recursion-on-Phenom, Insitro, and the broader academic community).

In practical terms, a paired landmark atlas plus one source-cell measurement is enough to start projecting target-cell transcriptomic responses for the same chemistry, with the strongest current confidence at the level of grouped chemical programs rather than individual molecules. The immediate use cases are counter-screen prioritization, cross-cell-line selectivity ranking, and decision support for which downstream cell types deserve direct profiling. The next dataset extends the same approach from in-library transfer to cross-library scaffold-family transfer in the same cell line, which is the closest analog to a fully prospective cross-cell-type prediction benchmark.

Methods

Data inputs

All analyses use the canonical Z-Screen RNA aggregate dataset, specifically:

- `canonical_obs.parquet`,
- `canonical_scvi_latents.parquet`,
- `canonical_chemistry.parquet` for grouped building-block benchmarks.

Only rows with a non-null `smiles_hash` (the public-bundle stand-in for a chemistry-resolved compound identity) and `chemistry_grain == "building_block_annotation"` were retained. Compound-control-only rows were not used for the transfer benchmark because their annotations are mechanism-of-action labels rather than chemistry coordinates.

scVI latent space rather than full gene expression

The transfer benchmarks are built on 32-dimensional scVI latent coordinates rather than full gene-level vectors, for two reasons. First, low-pass per-well RNA depth in the present dataset (each well averages 5 to 10 cells, sequenced as one pseudobulk row) means individual genes are dominated by sampling noise; scVI absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments require. The exploratory gene-level decoder experiment is included as a separate analysis specifically to assess how much of the latent gain survives when the result is lifted back to gene level.

Centered-response as the primary metric

Two cell lines have different baseline transcriptional states regardless of any perturbation. A cross-cell predictor that “knows” the constant baseline shift will show high raw-space cosine even if it has learned nothing about perturbational response. Centering removes that confound. For each train/test split, source and target training means were subtracted before fitting and evaluation. This is the closest available proxy for a vehicle-control normalization in a dataset that does not have matched vehicle controls.

Molecule-level transfer benchmark

For each (library, cell line, SMILES), scVI latent coordinates were averaged across wells. Shared molecules between a source and target cell line were then benchmarked under two evaluation spaces, `raw_profile` and `centered_response`. The models compared were `identity`, `global_shift` (raw space only), `ridge_transfer`, `pls_transfer`, and `knn_transfer`. The `global_shift` baseline was included so that the bulk-offset component of cross-cell variance is visible separately from chemistry-specific signal. Performance was measured by row-wise cosine similarity on held-out molecules and aggregated as mean cosine across multiple train-test splits.

Grouped building-block transfer

Within each library and cell-pair direction, molecules were grouped by candidate building-block keys (bb0_id, bb1_id, or multi-column combinations such as bb0_id+bb1_id). For each shared grouped key, source and target centered latent signatures were averaged and benchmarked using a centered ridge transfer model across repeated train-test splits. Performance was summarized as support-weighted cosine similarity across grouped programs. Support weighting prevents a few groups with thousands of supporting wells from being drowned out by many groups with five.

Cross-library transfer

For a fixed cell-pair direction, a centered ridge transfer matrix was trained in one library and applied to molecules from another library. Because the sparser libraries are highly singleton-heavy, adaptive minimum-well thresholds were used so that denser libraries retained stricter filters while still allowing exploratory sparse projections. The directional asymmetry between dense-to-sparse and sparse-to-dense transfer is a consequence of how informative each library is as a teacher; it is not corrected for in the metric.

Exploratory gene-level transfer decoder

For the four dense paired directions, held-out shared SMILES were evaluated with a composed latent-to-gene workflow. A centered ridge model first mapped source-cell scVI centroids into target-cell scVI space. Separately, a target-cell ridge decoder was trained on the training compounds to map target-cell scVI centroids to log-CPM pseudobulk deltas over 800 variable genes selected from target training cells. Held-out predictions were scored by per-compound Pearson and Spearman correlation over gene deltas and by top-50 / top-100 absolute-delta gene overlap.

Limitations

- Matched vehicle controls are unavailable in the canonical dataset, so the centered-response benchmark is the closest available proxy for perturbational response rather than a fully DMSO-normalized effect.
- The strongest molecule-level benchmark exists only for the denser paired libraries ZEL024 and ZEL031; ZEL028-2 becomes too sparse under a strict per-molecule support filter.
- Cross-library transfer is limited by sparse sampling and by disjoint building-block vocabularies across libraries, which prevents a clean cross-library tuple-transfer formulation today.
- All analyses are based on averaged 32-dimensional scVI latent signatures rather than full transcript-level models or time- and dose-aware transfer.
- The gene-level decoder experiment is exploratory and uses a target-cell decoder trained on observed target profiles; it should be read as a way to generate ranked gene hypotheses from transferred latents, not as evidence that full transcriptomes can already be prospectively reconstructed from an unmeasured target cell state.
- The current dataset reflects a small number of pilot experiments. The platform is engineered to scale by orders of magnitude beyond the present dataset.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper4/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper4/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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